

Question Number	Answer	Additional guidance	Mark
5(a)	An answer that includes three of the following points: • deforestation / loss of habitat (1) • increased grazing (of forest) (1) • reduced number of new trees (1) • disease (that has killed the trees) (1) • reduced genetic diversity (1)	ACCEPT loss of trees due to natural events e.g. landslides / tsunamis / fire / poor soil quality ACCEPT new grazing species e.g. reduced {pollination / reproduction / pollinators} / tree doesn't flower often / reduced enzyme activity due to {changing temperatures / drought} / few trees producing seeds / {air pollution / global warming / climate change} e.g. fungal disease ACCEPT reproduce asexually so no variation	(3)

Question Number	Answer	Additional guidance	Mark
5(b)(i)	• correct line measurement and conversion (1) • correct calculation (1) • correct answer in standard form (1)	ACCEPT $\pm 1\text{mm}$ <u>Example of calculation:</u> $38\,000\mu\text{m}$ $(38\,000 \div 200) = 190\,(\mu\text{m})$ $1.9 \times 10^2\,(\mu\text{m})$ ecf applies Correct answer with no working shown scores full marks	(3)

Question Number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes the following points: • {transport/ translocation} of {sucrose / amino acids} (1) • from {source /site of production / leaves} to {sink / site of use / site of storage} (1)	ACCEPT products of photosynthesis / lipids / hormones / mRNA / sugars ignore glucose	(2)

Question Number	Answer	Mark
5(b)(iii)	The only correct answer is C two <i>A is not correct because all the statements are correct apart from, they do not contain cytoplasm</i> <i>B is not correct because all the statements are correct apart from, they do not contain cytoplasm</i> <i>D is not correct because they do not contain cytoplasm</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(c)	An answer that includes three of the following points: - the edge of the xylem is closer to the {outside of the structure / epidermis} (than the phloem) in the root / the phloem is closer to the {outside of the structure / epidermis} than the xylem in the stem (1) - xylem and phloem are contained in (separate vascular) bundles in the stem (1) - more equal phloem:xylem in stem (whereas 4:1 in root) (xylem and phloem / (vascular) bundles) are arranged in {outer part of / a circular pattern in} stem (1)	ACCEPT xylem and phloem are separated by cambium in stem ignore sclerenchyma/parenchyma ACCEPT xylem and phloem together make an oval shape ignore {sizes/number unqualified} of {xylem/phloem} ACCEPT both xylem and phloem are closer to epidermis in stem ACCEPT (xylem and phloem/ (vascular) bundles) are arranged in all areas of stem (in monocots) ACCEPT vascular tissue in one area in the root whereas in several areas in the stem ACCEPT labelled diagram showing the distribution in the stem	(3)